Marsabit County AI Complaint Management System

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CHAPTER 1: INTRODUCTION

Introduction

* Digital transformation is critical in modern governance
* Efficient complaint handling improves service delivery
* Marsabit County faces:
  * Vast geographical coverage
  * Limited access to government services
* Need for an **intelligent, automated system**

Background of the Study

* Current systems:
  * Manual and paper-based
* Problems:
  * Delays
  * Data loss
  * Lack of transparency
* AI enables automation and efficiency

Current / Existing System

* Physical complaint submission
* Suggestion boxes
* Manual routing
* Informal communication channels

Problem Statement

* Complaints are:
  * Lost or delayed
  * Not tracked
* Poor coordination between departments
* No centralized digital system

Proposed System

* AI-powered complaint management system
* Supports:
  * Text
  * Image
  * Audio
* Automates classification, routing, prioritization

Objectives & Justification

General Objective

* Develop an AI-based complaint management system

Specific Objectives

* Design system architecture
* Develop AI models
* Automate complaint handling
* Improve transparency

Justification

* Improves efficiency
* Enhances transparency
* Builds trust

Scope, Significance & Limitations

Scope

* Marsabit County
* Multi-user system

Significance

* Better service delivery
* Data-driven decisions

Limitations

* Internet connectivity
* Digital literacy
* Language diversity

CHAPTER 2: LITERATURE REVIEW

Literature Review

* Review of:
  * Complaint management systems
  * AI in governance
  * NLP and ML models

Existing Systems & Gaps

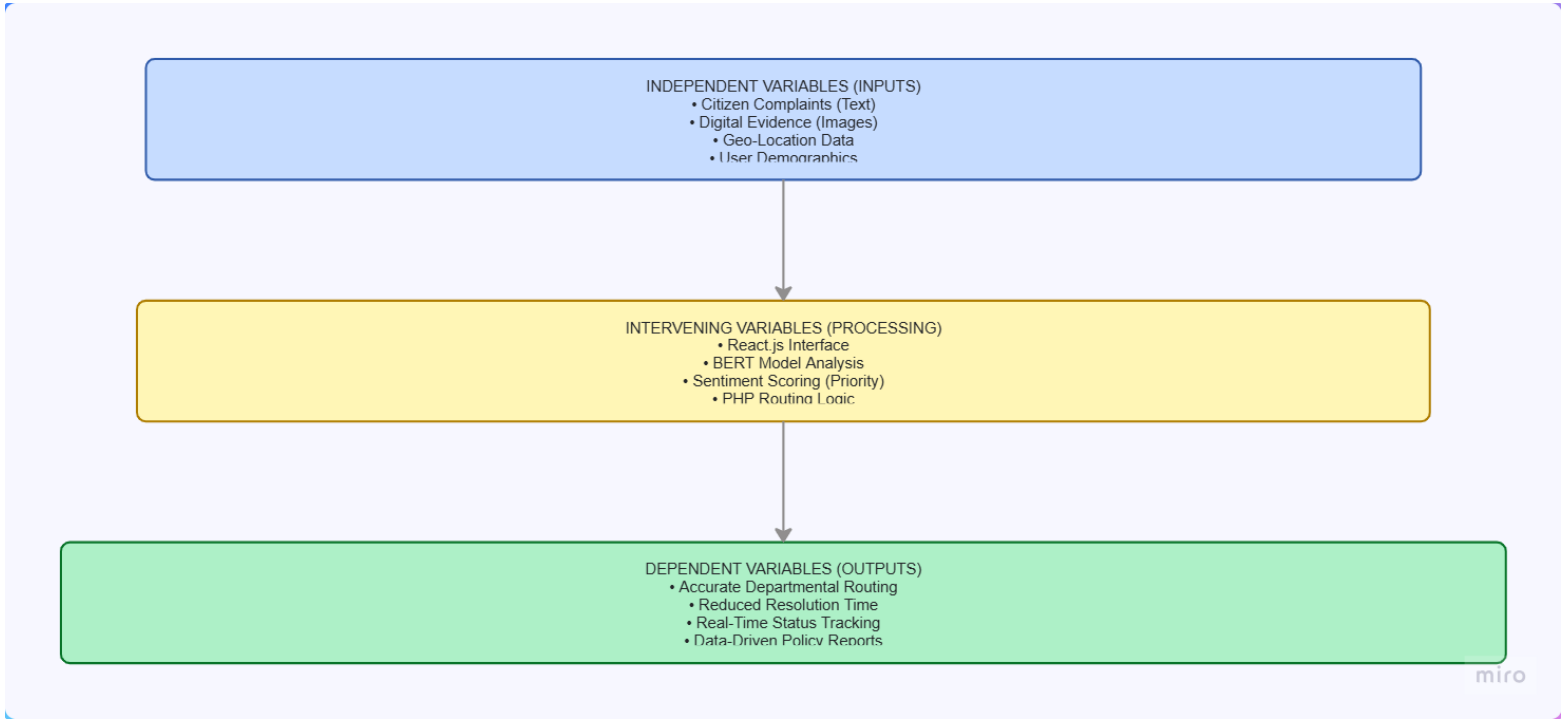
* Manual systems → inefficient
* Digital systems → limited intelligence
* Gaps:
  * No automation
  * No AI classification
  * Poor scalability

Table 2. 1: Comparison of manual systems, existing complaints systems and proposed AI system

Feature	Manual System (Traditional)	Existing Digital Systems (e.g., Email/Static Web Forms)	Proposed AI System (County AI Complaint Management)
Submission Channel	Physical suggestion boxes, letters, walk-in offices.	Standard web forms, emails, or social media tags.	**React.js Web Portal** (Responsive, offline-capable, & Interactive).
Routing Mechanism	Manual sorting by a secretary or administrator.	Manual forwarding by a backend IT officer.	**Automated BERT Model** (Instant classification by departmental context).
Prioritization	First-In, First-Out (FIFO). No urgency detection.	Manual tagging. Often ignores urgency unless flagged by a human.	**Sentiment Analysis** (AI detects distress/urgency and auto-flags high priority).
Feedback Loop	None or very slow (Physical letters).	Email auto-reply (often generic). No real-time tracking.	**Real-Time Dashboard** (Live status updates & "Push" notifications for citizens).
Data Analytics	None. Paper records are hard to analyze or aggregate.	Basic counts (e.g., "50 emails received").	**Deep Analytics** (Heatmaps of problem wards, longitudinal trend analysis).
Turnaround Time	**High** (Weeks to Months due to physical logistics).	**Medium** (Days to Weeks; still reliant on human triage).	**Low** (Minutes for routing; Hours for initial officer response).

Conceptual Framework

* Input → Processing → Output
* AI acts as decision engine


INDEPENDENT VARIABLES (INPUTS)
• Citizen Complaints (Text)
• Digital Evidence (Images)
• Geo-Location Data
• User Demographics
INTERVENING VARIABLES (PROCESSING)
• React.js Interface
• BERT Model Analysis
• Sentiment Scoring (Priority)
• PHP Routing Logic
DEPENDENT VARIABLES (OUTPUTS)
• Accurate Departmental Routing
• Reduced Resolution Time
• Real-Time Status Tracking
• Data-Driven Policy Reports
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CHAPTER 3: METHODOLOGY

Research Methodology

* Agile development model
* Iterative development
* Continuous improvement

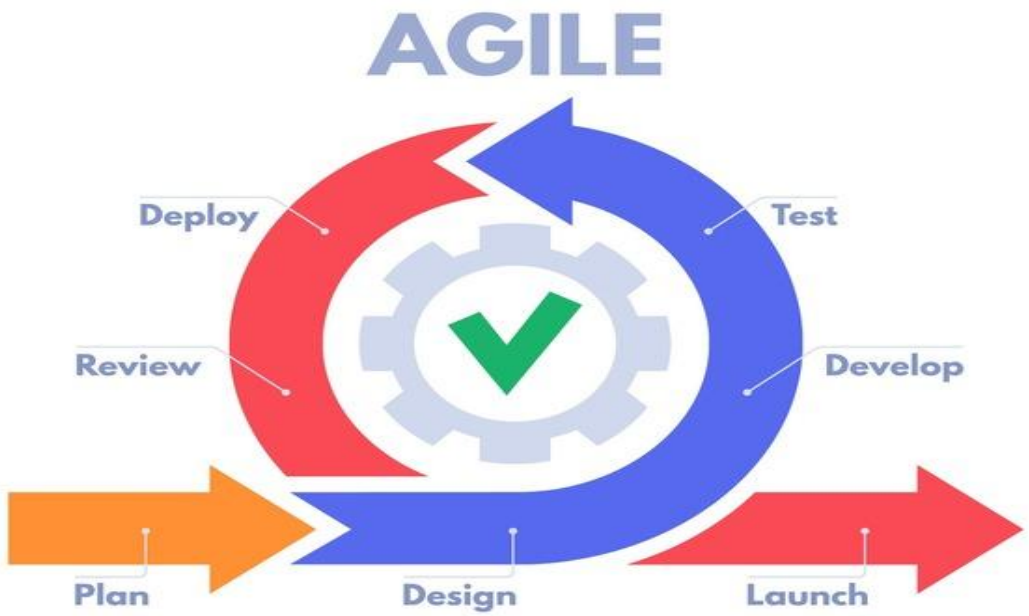


Figure 3.1 scrum process

Data Acquisition

* No existing dataset available
* Created custom dataset
* Sources:
  * Simulated complaints
  * Government categories
  * Real-world scenarios

Multi-Modal Data Sources

* Text complaints
* Image data
* Audio data (voice complaints)

Data Preprocessing

* Text cleaning & tokenization
* Image normalization
* Audio → text (Whisper)
* Label encoding

AI Models Used

* BERT → text classification
* Computer Vision → image analysis
* Whisper → speech-to-text
* Zero-shot → flexible classification

AI Training Pipeline

* Data collection
* Preprocessing
* Training
* Validation
* Deployment

CHAPTER 4: SYSTEM DESIGN

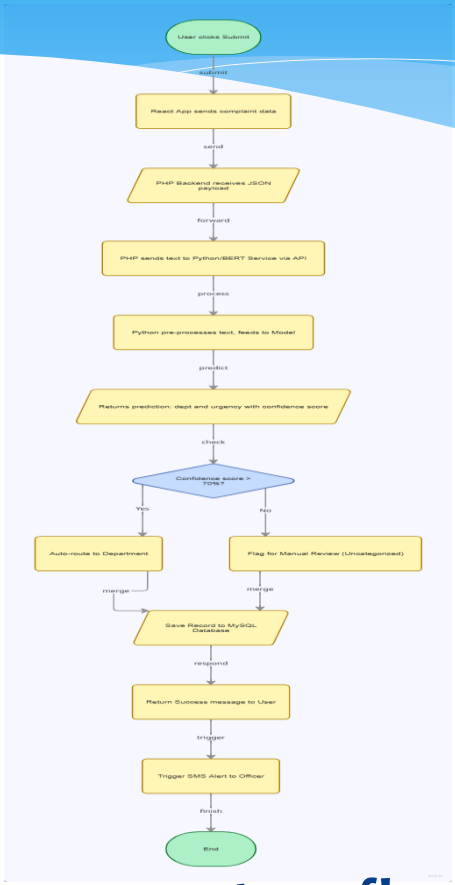
System Architecture

* Frontend: React.js
* Backend: PHP
* Database: MySQL
* AI Layer integrated

Figure 4. 1 : use case diagram

System Process Flow

* Complaint submission → AI processing → routing → feedback



* **Figure 4. 2: system flowchart**

Database Design

* Tables:
  * Users
  * Complaints
  * Departments
  * Status

Table 4.1 : complaints

Field Name	Data Type	Key	Description
id	INT (11)	Primary Key (PK)	Auto-incrementing unique identifier for each complaint.
ticket_ref	VARCHAR (20)	Unique	Tracking ID used to identify complaints (e.g., MST-2025-001).
sub_county	VARCHAR (30)	Index	Specific region where the complaint originates (e.g., North Horr).
ward	VARCHAR (30)	—	Specific ward within the sub-county (e.g., Illeret).
complaint_text	TEXT(50)	—	Detailed description of the issue submitted by the citizen.
predicted_dept	VARCHAR (100)	—	Department automatically determined by the AI system.
sentiment_score	FLOAT	—	AI-generated urgency score ranging from 0.0 to 1.0.
status	ENUM	—	Complaint processing status: Pending, In Progress, or Resolved.
created_at	DATE	—	Date and time when the complaint was submitted.

CHAPTER 5: TESTING & IMPLEMENTATION

Testing & Implementation

* Testing:
  * Unit testing
  * Integration testing
  * System testing
* Tools:
  * React, PHP, MySQL
* Deployment:
  * Server setup

Challenges Encountered

* Limited dataset availability
* Internet connectivity issues
* Model accuracy limitations
* Integration complexity (AI + system)

CHAPTER 6: CONCLUSION & RECOMMENDATIONS

Conclusion

* Manual systems are inefficient
* AI improves:
  * Speed
  * Accuracy
  * Transparency
* System enhances governance

Recommendations & Future Work

Recommendations

* Deploy system county-wide
* Train users
* Improve infrastructure

Future Work

* Integrate SMS/USSD
* Improve multilingual support
* Continuous AI model training
* Real-time analytics enhancement

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